XRPL Validator — Daily Engineering Report · 2026-06-14

Period: 2026-06-14 00:00 to 24:00 US Eastern (2026-06-14 04:00 to 2026-06-15 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 7 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-06-14): 0.51504 (10,515 missed of 21,682). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(validator): remove dead full-DB sweep footgun (reaudit F2)** (`a99dbb1`, xrpl-validator) start_sweep + fetch_batch re-fetched every state.rocks key at one pinned ledger and wrote each with an unconditional db.put NOT paired with a hasher update. Run alongside live sync it regressed freshly-synced keys to stale pinned-seq values while the in-memory hasher kept the live value — silent rocks rot that detonates on the next restart (hasher rebuilt from rocks → wrong account_hash → 3 strikes → halt). Zero …
- **fix(validator): refuse signing startup with untrusted RPC endpoints (reaudit F7)** (`7501eab`, xrpl-validator) A signing validator on a host with no local rippled (the validator host) silently fails over from localhost to the public clusters (xrplcluster.com / s1.ripple.com) when XRPL_RPC_URL/XRPL_WS_URL are unset — a consensus-divergence source and a violation of the never-hammer-public-RPCs rule. The binary never enforced the operational 'MUST set both env vars' rule.
- **feat(validator): encode crash-consistency contract via clean-shutdown marker (reaudit F5)** (`28f6658`, xrpl-validator) The 'wipe state.rocks + bulk-resync on unclean restart' contract lived only in operator discipline. A clean shutdown now writes clean_shutdown.marker (SIGTERM handler, after the leaf-cache save); boot consumes it. Marker present => prior shutdown clean, state consistent, resume. Marker absent while state.rocks exists => unclean (OOM/SIGKILL/power) => state may be torn (hasher rebuilds from inconsistent rocks → …
- **feat(validator): refresh active amendments on flag ledgers (reaudit F6/va-07)** (`ecf3ebc`, xrpl-validator) The FFI verifier fetched the active mainnet amendment set once at startup and froze it for the process lifetime. An amendment that reaches 2-week majority and activates while the validator runs would be absent → every tx exercising it diverges → 3 strikes → halt; this fires more often in the post-3.2.0 wave.

- **fix(ledger): preserve leaf data on SHAMap delete collapse (reaudit F4 / val-110)** (`826406e`, xrpl-validator) delete_from_mut's structural-collapse path promoted the surviving leaf via find_sole_leaf -> LeafNode::new_hash_only(key, hash), which zeroes data (data: Vec::new()). In a State tree the surviving sibling kept its hash but lost its bytes — invisible to the hash-based signing path, but lookup() returned Some([]). This blocked lockstep M1 and any path trusting SHAMap::delete.

Docs and operations

- **chore(validator): commit ffi_test sidecar arg drift to extended apply_ledger_in_order signature (reaudit F8a)** (`de6ea4b`, xrpl-validator)
- **style(validator): use u32::is_multiple_of in is_flag_ledger (clippy)** (`2279891`, xrpl-validator)

Independence program

No independence record for UTC day 2026-06-14; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
06-14 13:10	—	xrpl-validator	`de6ea4b`	chore(validator): commit ffi_test sidecar arg drift to extended apply_ledger_in_order signature (reaudit F8a)
06-14 13:21	—	xrpl-validator	`a99dbb1`	fix(validator): remove dead full-DB sweep footgun (reaudit F2)
06-14 13:28	—	xrpl-validator	`7501eab`	fix(validator): refuse signing startup with untrusted RPC endpoints (reaudit F7)

Committed	Merged into main	Repo	Commit	Subject
06-14 13:34	—	xrpl-validator	`28f6658`	feat(validator): encode crash-consistency contract via clean-shutdown marker (reaudit F5)
06-14 13:47	—	xrpl-validator	`ecf3ebc`	feat(validator): refresh active amendments on flag ledgers (reaudit F6/va-07)
06-14 13:53	—	xrpl-validator	`826406e`	fix(ledger): preserve leaf data on SHAMap delete collapse (reaudit F4 / val-110)
06-14 14:10	—	xrpl-validator	`2279891`	style(validator): use u32::is_multiple_of in is_flag_ledger (clippy)

Times in UTC. A commit appears on the day it was committed to main or merged into it.